Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	27 January 2026
Team ID	LTVIP2026TMIDS79347
Project Name	Online payment fraud detection using machine learning
Maximum Marks	4 Marks

Technical Architecture:

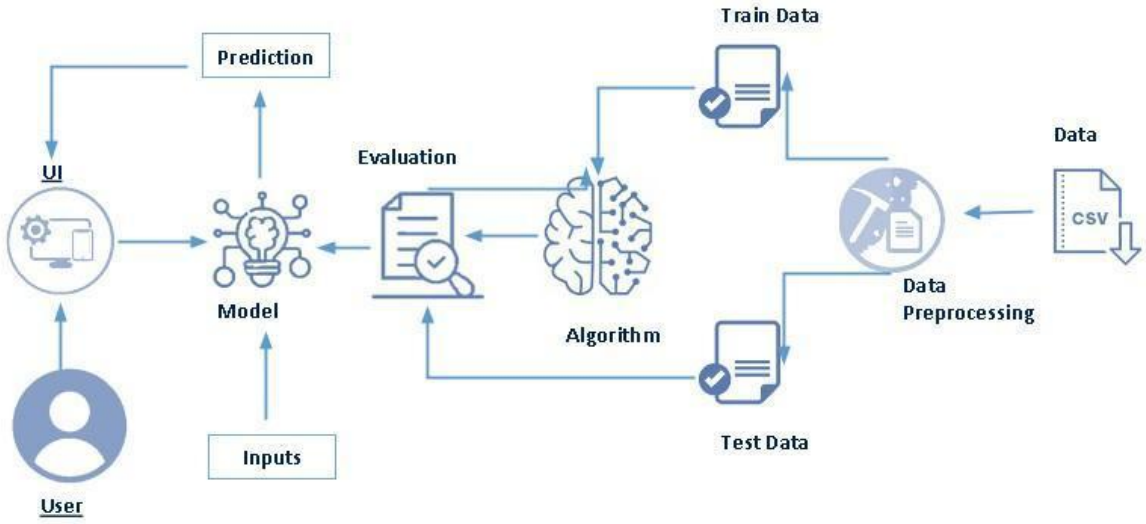

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web interface where users enter transaction details and view fraud prediction results	HTML, CSS, JavaScript, Bootstrap
2	Application Logic-1	Backend logic to handle request/response between UI and ML model	Python, Flask
3	Application Logic-2	Data preprocessing, feature engineering, and transformation	Pandas, NumPy, Scikit-learn
4	Application Logic-3	Fraud prediction logic using trained ML model	Random Forest / XGBoost / Logistic Regression
5	Database	Stores transaction data and user logs	MySQL / PostgreSQL
6	Cloud Database	Cloud-based storage for scalable transaction records	AWS RDS / MongoDB Atlas
7	File Storage	Stores trained ML model file and datasets	Local Storage / AWS S3
8	External API-1	Payment gateway transaction integration	Razorpay API / Stripe API
9	External API-2	User authentication and identity verification	Aadhaar API / OTP Verification API
10	Machine Learning Model	Detect fraudulent transactions based on patterns	Random Forest Classifier / XGBoost
11	Infrastructure (Server / Cloud)	Application deployment and hosting	AWS EC2 / Heroku / Docker / Kubernetes

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for development	Flask, Scikit-learn, Pandas
2.	Security Implementations	Secure transactions and prevent attacks	HTTPS, SSL/TLS, SHA-256, JWT Authentication, Firewalls
3.	Scalable Architecture	3-Tier architecture (Presentation, Application, Data layer)	Flask + REST API + Cloud DB
4. 4	Availability	High availability using load balancing & cloud hosting	AWS Load Balancer, Cloud Deployment